Database Concepts - Assignment 4

SQL Queries

(Task D)

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Question 1: For any two given months (i.e., you can assume any two months, e.g., April 2022 and May 2022), list the months, the total number of vaccines administered in each observation months in each of all countries, and the difference between the administered vaccines. Each row in the result set must have the following structure. (Note: OM2 is after OM1.

Query

```
SELECT
'January 2021' AS "Observation Months 1 (OM1)",
UPPER(L.CountryName) AS "Country Name",
UPPER(OM1.total_vaccinations) AS "Administered Vaccine on OM1(VOM1)",
'December 2021' AS "Observation Months 2 (OM2)",
UPPER(OM2.total_vaccinations) AS "Administered Vaccine on OM2(VOM2)",
UPPER(OM1.total_vaccinations) - UPPER(OM2.total_vaccinations) AS "Difference of totals(VOM1 -
VOM2)"
FROM
(SELECT ISO_Code, SUM(total_vaccinations) AS total_vaccinations
FROM Vaccination
WHERE Date BETWEEN '01-01-2021' AND '31-01-2021'
GROUP BY ISO_Code) AS OM1
JOIN
(SELECT ISO_Code, SUM(total_vaccinations) AS total_vaccinations
FROM Vaccination
WHERE Date BETWEEN '01-12-2021' AND '31-12-2021'
GROUP BY ISO_Code) AS OM2
ON OM1.ISO_Code = OM2.ISO_Code
JOIN
Location AS L
ON OM1.ISO_Code = L.ISO_Code;
```

Grid view | Form view

Total rows loaded: 223

	Observation Months 1 (OM1)	Country Name	Administered Vaccine on OM1(VOM1)	Observation Months 2 (OM2)	Administered Vaccine on OM2(VOM2)	Difference of totals(VOM1 - VOM2)
1	January 2021	ARUBA	98520619.0	December 2021	97733163.0	787456
2	January 2021	AFGHANISTAN	1640479756.0	December 2021	1598823216.0	41656540
3	January 2021	ANGOLA	1397580051.0	December 2021	1372509596.0	25070455
4	January 2021	ANGUILLA	1290079.0	December 2021	1271375.0	18704
5	January 2021	ALBANIA	386070141.0	December 2021	378451500.0	7618641
6	January 2021	ANDORRA	5705810.0	December 2021	5606210.0	99600
7	January 2021	UNITED ARAB...	5541037533.0	December 2021	5500822883.0	40214650
8	January 2021	ARGENTINA	91986019398	December 2021	90795181962	1190837436
9	January 2021	ARMENIA	57891991.0	December 2021	56295603.0	1596388
10	January 2021	ANTIGUA AND...	7204949.0	December 2021	7293302.0	-88353

Figure 1: Output for Task D.1

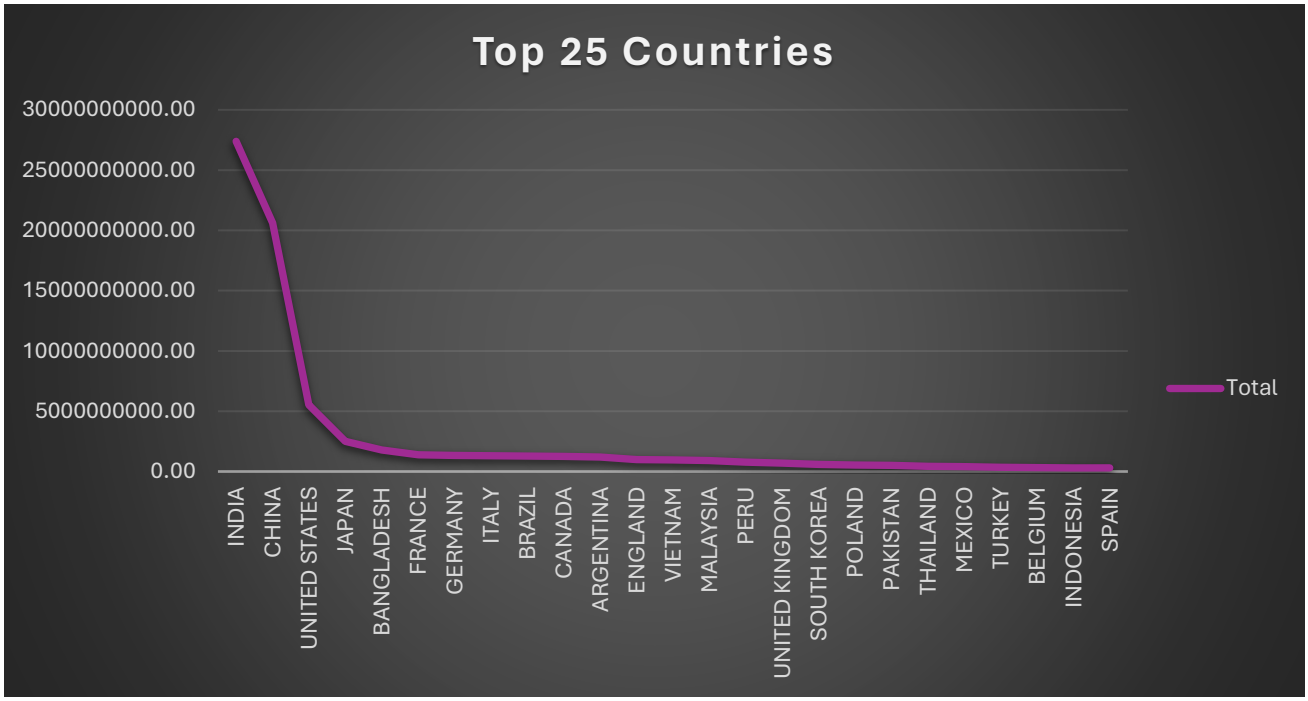


Figure 2: Top 25 countries record visualisation with the highest cumulative values

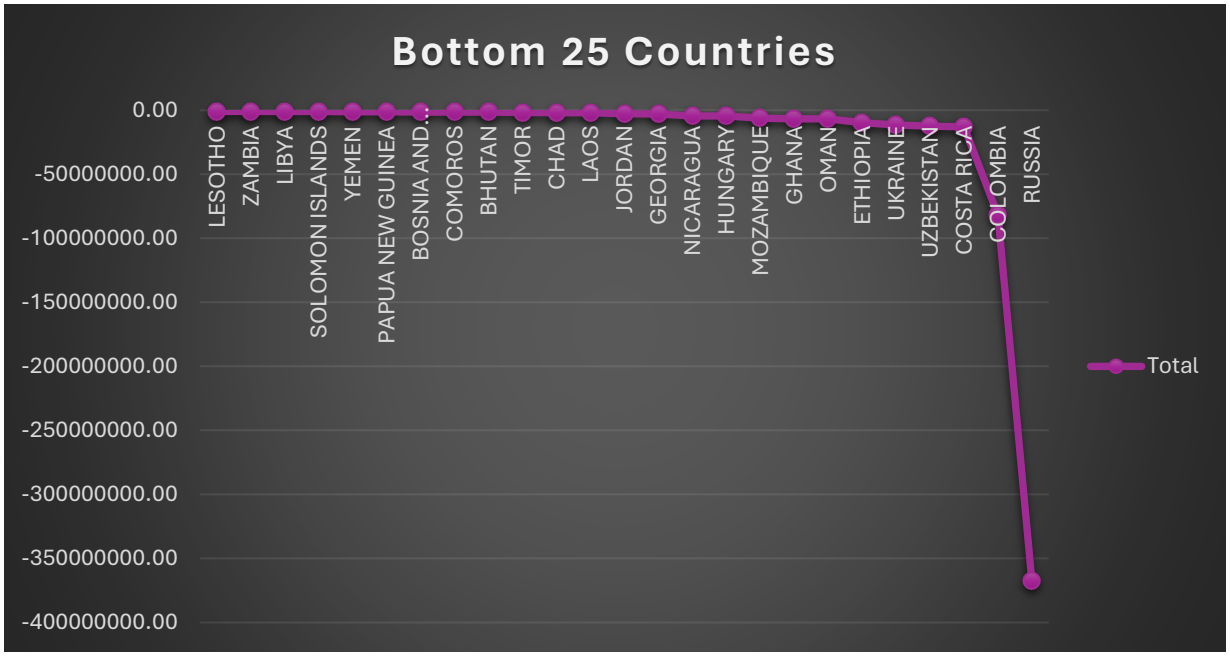


Figure 3: Bottom 25 countries record visualisation with the lowest cumulative values

*Negative values represent that more people were vaccinated in OM2 than in OM1 in these countries.

Question 2: Find the countries with more than the average cumulative numbers of COVID-19 doses administered by each country in each month (Note: the result may include multiple countries or a single country). Produces a result set containing the name of each country and the cumulative number of doses administered in that country. Each row in the result set must have the following structure.

Query

```
SELECT
UPPER(MC.CountryName) AS "Country Name",
strftime('%m/%Y', datetime(substr(V.Date, 7, 4) || '-' || substr(V.Date, 4, 2) || '-' || substr(V.Date, 1, 2), 'start of month')) AS "Months",
SUM(V.total_vaccinations) AS "Cumulative Doses"
FROM
Vaccination V
JOIN
Location MC ON V.ISO_Code = MC.ISO_Code
WHERE
(
SELECT
SUM(V2.total_vaccinations)
FROM
Vaccination V2
) > (
SELECT
AVG(V2.total_vaccinations)
FROM
Vaccination V2
)
GROUP BY
MC.ISO_Code;
```

Grid view | Form view

Total rows loaded: 223

	Country Name	Months	Cumulative Doses
1	ARUBA	03/2021	100298978
2	AFGHANISTAN	02/2021	1682111829
3	ANGOLA	03/2021	1426308398
4	ANGUILLA	02/2021	1290079
5	ALBANIA	01/2021	395364296
6	ANDORRA	01/2021	5846260
7	UNITED ARAB EMIRATES	01/2021	5667655630
8	ARGENTINA	12/2020	93762123135
9	ARMENIA	03/2021	58724412
10	ANTIGUA AND BARBUDA	02/2021	7482788

Figure 4: Output snapshot for Task D.2

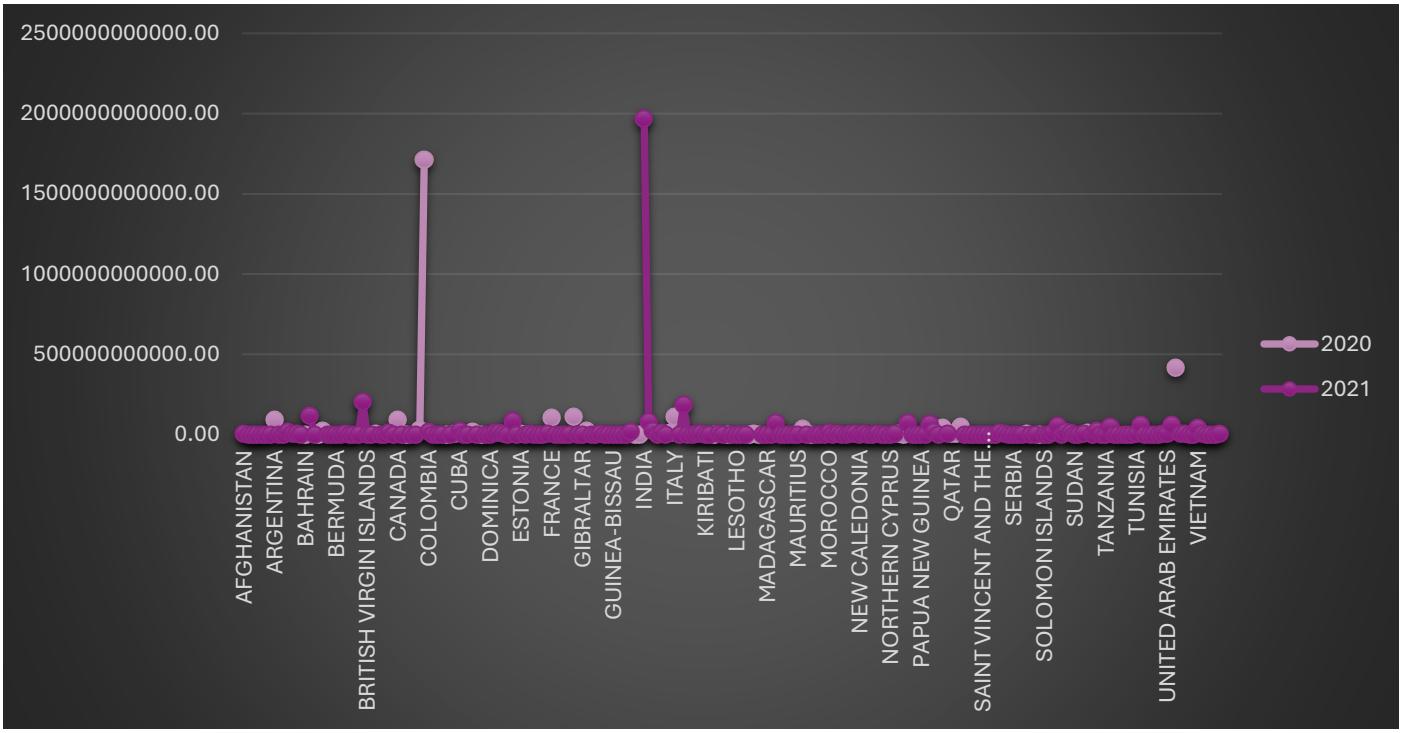


Figure 5: Visual representation of all countries and their Cumulative Doses for years 2021 and 2022

Question 3: Produce a list of vaccine types with the countries taking each vaccine type. For a vaccine type that has been taken in multiple countries, the result set is required to show several tuples reporting each country in a separate tuple. Each row in the result set must have the following structure.

Query

```
SELECT
UPPER(V.Name) AS "Vaccine Type",
UPPER(L.CountryName) AS "Country"
FROM
Uses_in U
JOIN
Vaccine V ON U.VaccineName = V.Name
JOIN
Location L ON U.ISO_Code = L.ISO_Code
ORDER BY
V.Name, L.CountryName;
```

Grid view | Form view

Total rows loaded: 957

	Vaccine Type	Country
1	ABDALA	CUBA
2	ABDALA	NICARAGUA
3	ABDALA	VENEZUELA
4	ABDALA	VIETNAM
5	COVIRAN BAREKAT	IRAN
6	CANSINO	AFGHANISTAN
7	CANSINO	ARGENTINA
8	CANSINO	BAHRAIN
9	CANSINO	CHILE
10	CANSINO	CHINA

Figure 6: Output snapshot for Task D.3

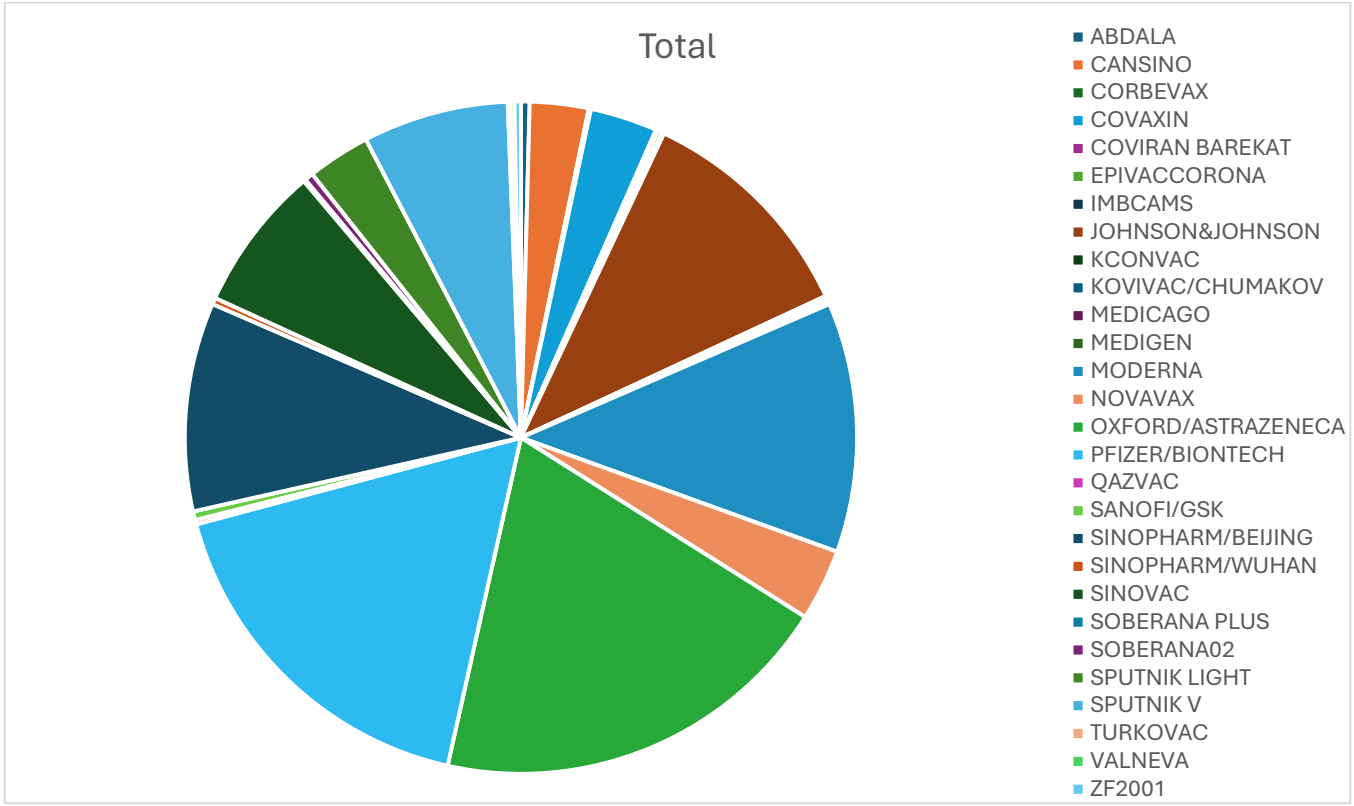
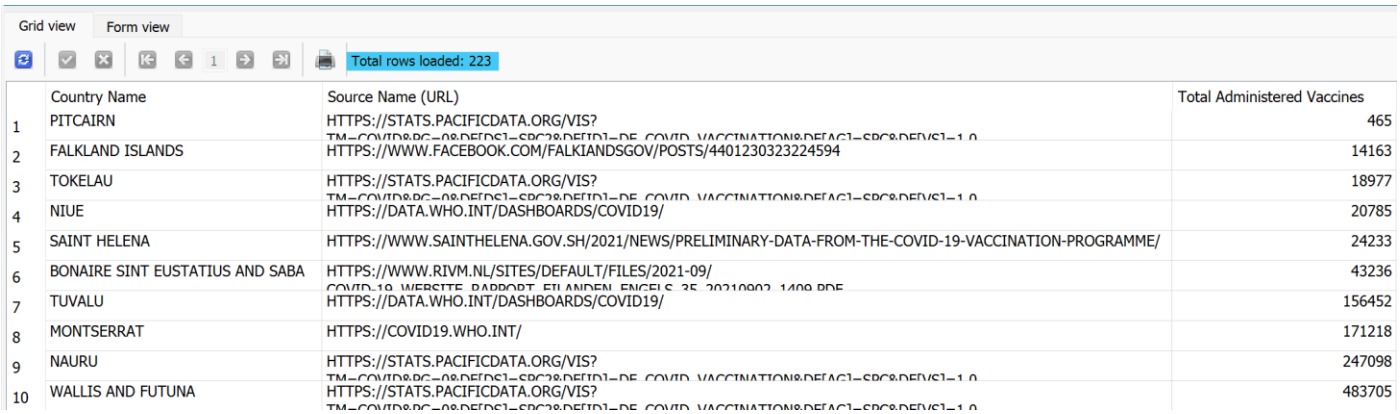


Figure 7: Pie chart depicting the vaccine proportions

Question 4: There are different data sources used to produce the dataset. Produce a report showing the total number of vaccines administered in each country according to each data source (i.e., each unique URL). Order the result set by the total number of administered vaccines. Each row in the result set must have the following structure.

Query

```
SELECT
UPPER(L.CountryName) AS "Country Name",
UPPER(L.SourceURL) AS "Source Name (URL)",
SUM(V.total_vaccinations) AS "Total Administered Vaccines"
FROM
Location L
JOIN
Vaccination V ON L.ISO_Code = V.ISO_Code
GROUP BY
L.CountryName
ORDER BY
SUM(V.total_vaccinations);
```



Grid view | Form view

Total rows loaded: 223

	Country Name	Source Name (URL)	Total Administered Vaccines
1	PITCAIRN	HTTPS://STATS.PACIFICDATA.ORG/VIS?	465
2	FALKLAND ISLANDS	HTTPS://WWW.FACEBOOK.COM/FALKLANDSGOV/POSTS/4401230323224594	14163
3	TOKELAU	HTTPS://STATS.PACIFICDATA.ORG/VIS?	18977
4	NIUE	HTTPS://DATA.WHO.INT/DASHBOARDS/COVID19/	20785
5	SAINT HELENA	HTTPS://WWW.SAINTHELENA.GOV.SH/2021/NEWS/PRELIMINARY-DATA-FROM-THE-COVID-19-VACCINATION-PROGRAMME/	24233
6	BONAIRE SINT EUSTATIUS AND SABA	HTTPS://WWW.RIVM.NL/SITES/DEFAULT/FILES/2021-09/	43236
7	TUVALU	HTTPS://DATA.WHO.INT/DASHBOARDS/COVID19/	156452
8	MONTSERRAT	HTTPS://COVID19.WHO.INT/	171218
9	NAURU	HTTPS://STATS.PACIFICDATA.ORG/VIS?	247098
10	WALLIS AND FUTUNA	HTTPS://STATS.PACIFICDATA.ORG/VIS?	483705

Figure 8: Output Snapshot for Task D.4

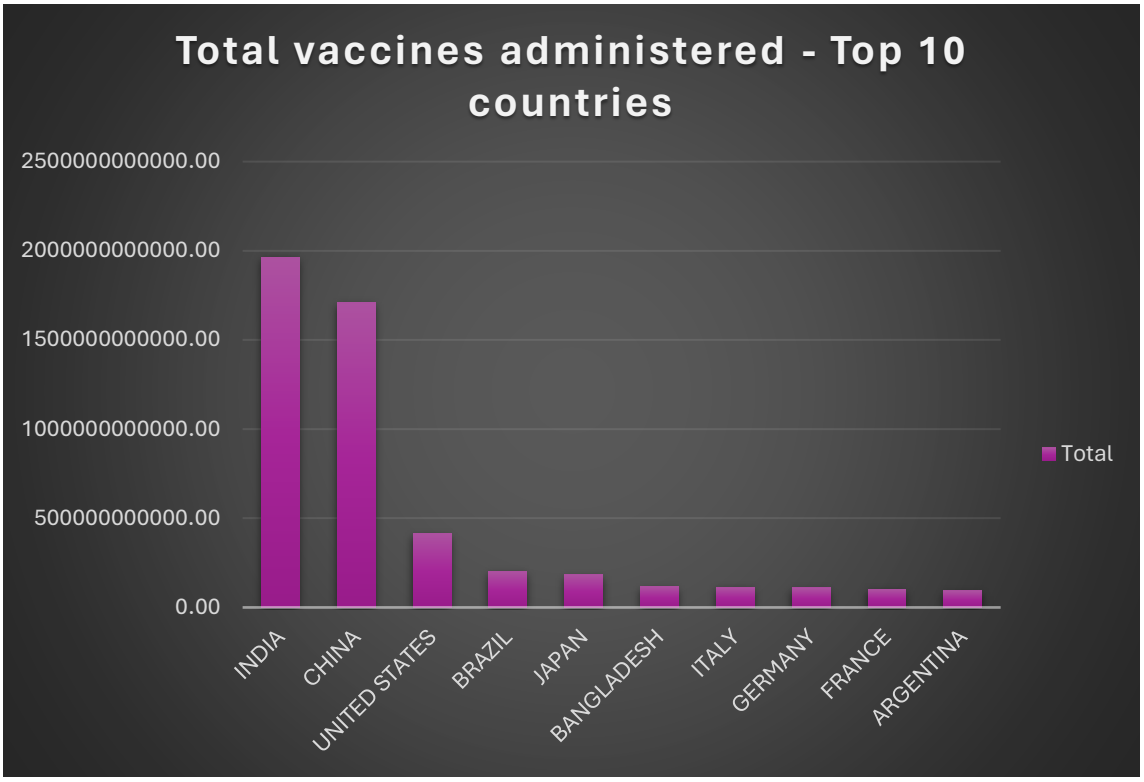


Figure 9: Bar chart depicting the top 10 countries and the total vaccinations administered by them

Question 5: How do various countries compare in the speed of their vaccine administration? Produce a report that lists all the observation months in 2022 and 2023, and then for each months, list the total number of people *fully vaccinated* in each one of the 4 countries used in this assignment.

Query

```
SELECT
strftime('%m/%Y', datetime(substr(V.Date, 7, 4) || '-' || substr(V.Date, 4, 2) || '-' || substr(V.Date, 1, 2), 'start of month')) AS "Date Range (Months)",
SUM(CASE WHEN L.CountryName = 'United States' THEN V.people_fully_vaccinated ELSE 0 END) AS "United States",
SUM(CASE WHEN L.CountryName = 'Wales' THEN V.people_fully_vaccinated ELSE 0 END) AS "Wales",
SUM(CASE WHEN L.CountryName = 'Canada' THEN V.people_fully_vaccinated ELSE 0 END) AS "Canada",
SUM(CASE WHEN L.CountryName = 'Denmark' THEN V.people_fully_vaccinated ELSE 0 END) AS "Denmark"
FROM
Vaccination V
JOIN
Location L ON V.ISO_Code = L.ISO_Code
WHERE
date(substr(V.Date, 7, 4) || '-' || substr(V.Date, 4, 2) || '-' || substr(V.Date, 1, 2)) BETWEEN '2022-01-01' AND '2023-12-31'
GROUP BY
```

```
strftime('%m/%Y', datetime(substr(V.Date, 7, 4) || '-' || substr(V.Date, 4, 2) || '-' || substr(V.Date, 1, 2), 'start of month'))
ORDER BY
strftime('%Y/%m', datetime(substr(V.Date, 7, 4) || '-' || substr(V.Date, 4, 2) || '-' || substr(V.Date, 1, 2), 'start of month'));
```

Grid view | Form view

Total rows loaded: 24

	Date Range (Months)	United States	Wales	Canada	Denmark
1	01/2022	6600697602	72520299	921332759	18550648
2	02/2022	6077229508	66583095	857165749	18751187
3	03/2022	6795486776	74219507	962730585	18785873
4	04/2022	6621916071	72215488	843766566	23500111
5	05/2022	6884154907	72524729	972005756	18810542
6	06/2022	6693229825	72722039	817146990	18820420
7	07/2022	6942374319	75530280	818313846	23535589
8	08/2022	6972177316	75960111	977097794	18832559
9	09/2022	6774563616	73720387	946890386	23544253
10	10/2022	7044514583	76343178	979956004	18838458

Figure 10: Output snapshot for Task D.5

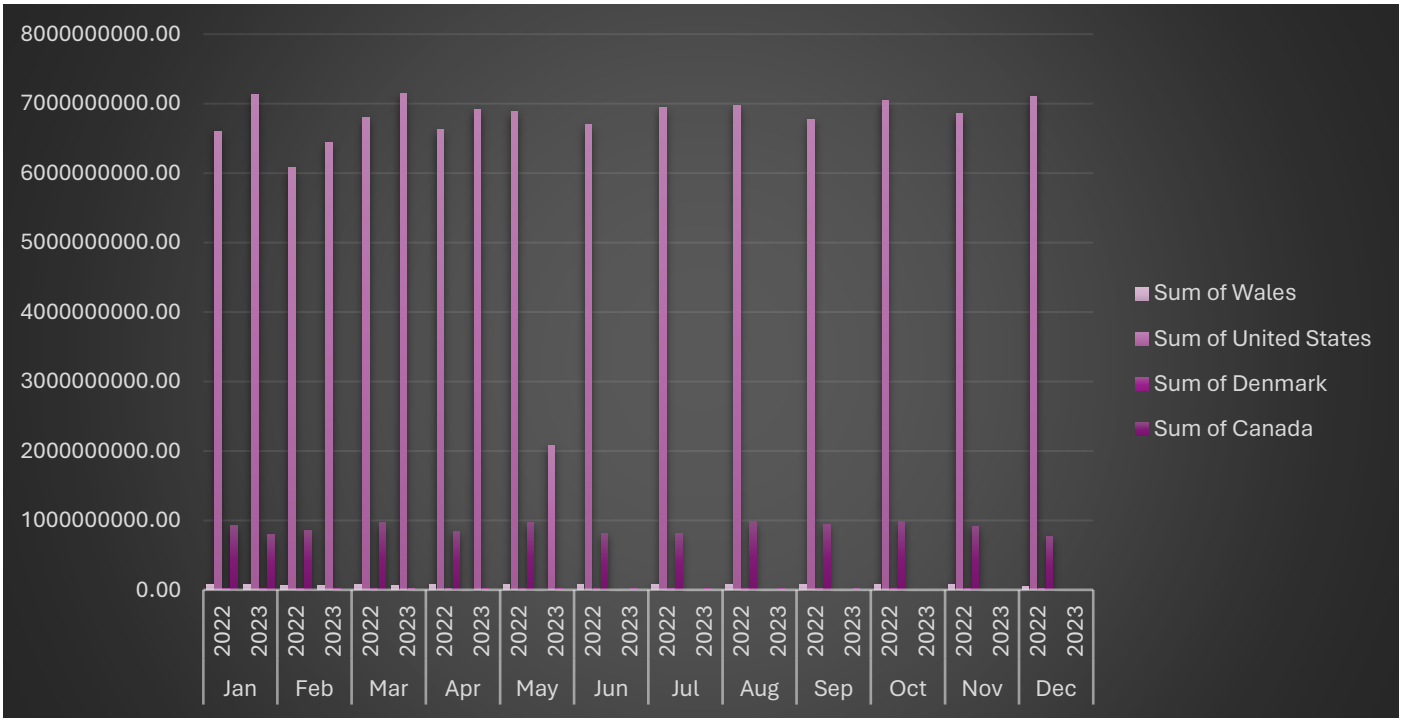


Figure 11: Bar chart depicting the monthly speed of vaccinations administered by the given 4 countries in the years 2022 and 2023

References

Javatpoint. https://www.javatpoint.com/substr-function-in-sql. Accessed on 14/06/2024.

IBM. https://www.ibm.com/docs/en/informix-servers/14.10?topic=expression-concatenation-operator. Accessed on 14/06/2024.

Dr Zhou, Xiangmin(Emily). (2024). https://rmit.instructure.com/courses/124806/modules. Canvas.